Data collection campaign: building a smell cartography

Cartography of Grenoble, Saint Martin d'Hères and Echirolles

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1. Introduction

For this project we propose the development of a dynamic cartography of odors, meaning that the displayed values can change over time as new data is updated, added, or deleted. We are assuming a system that updates hourly.

The odors considered for this map are divided in 5: coffee aroma, floral, trash, food aroma, fumes. The objective of this map is to inform users of these aromas by location and intensity with a color schema, and to particularly focus on unpleasant smells (trash and fumes) to help users to avoid them. This in contrast to the example maps provided in class (Scentscape Singapore and The Scents & Smells of Paris) which has a touristic focus.

2. Table

Type of data	*Associated V's*	*Contribution*	*Strategies for collection*
Citizen smell reports	Variety, Value	Captures subjective human perception of smells and their emotional impact, helping define how residents experience the city's olfactory character regarding a specific street.	Mobile/web app for users to log smell type, intensity, and time with GPS tagging. An example could be the use of data from OdourCollect App.
Environmental sensor data NO2, and CO2	Volume, Velocity	Provides objective chemical measures linked to urban smells, showing pollution or freshness levels in different areas.	Use existing city air-quality sensors; deploy portable sensors on buses, bikes etc.
Weather and climate data (humidity, wind, temperature)	Velocity, Variety	Can be used to explain how climate conditions affect smell intensity and spread across different times of day or seasons	Gather data from open weather APIs For example OpenWeather.
Location of restaurants and parks	Volume, Variety, Value	Helps identify scent hotspots by linking geographic locations to potential smell sources such as food aromas in restaurants or floral smells in parks.	From Google Maps API, OpenStreetMap using API querying.

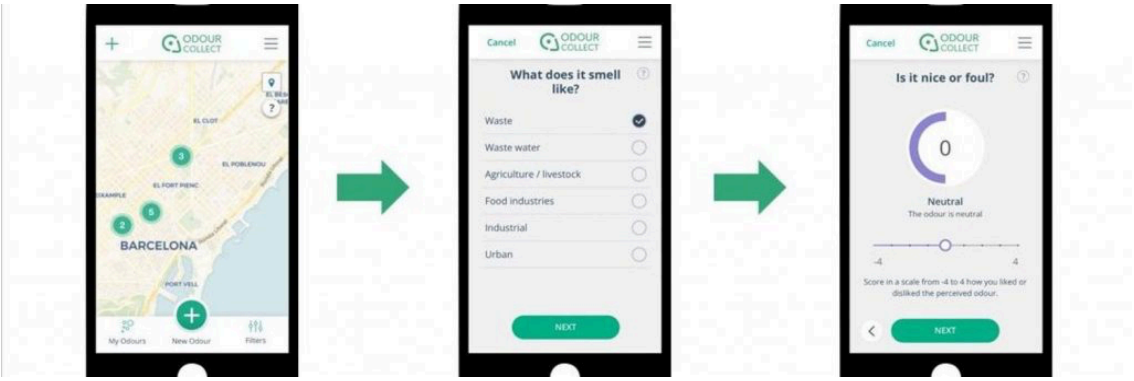
Type of data	*Associated V's*	*Contribution*	*Strategies for collection*
Traffic	Velocity, Volume	Correlates emission-related smells (fumes) with traffic patterns and crowd activity	Access city mobility open data of patterns. For example TomTom Traffic Index
Location of waste disposal and collection points	Variety, Value	Helps identify odor hotspots related to trash.	From open data portals of Grenoble Alpes Métropole: Waste collection and treatment points and voluntary contribution points and Metropolitan area waste disposal sites

3. Discussion:

The following section focuses on how each data identified can contribute to build the cartography with the goal established in the introduction.

CITIZEN SMELL REPORTS

By the extraction of data relative to qualitative and quantitative description of odors in specific streets, we can have specific points in our map that can be used also for another analysis (see in the part of weather and climate data) to have a more accurate descriptions of odor that could not be easily identified with the area focus data. An app we found that can be used for this data is OdourCollect, a tool based on a 'bottom-to-top' approach to address odor pollution and empower citizens through citizen science. However, one consideration with this data is maintaining the newer information, 1 to 3 months old can be considered, as the older information can affect the veracity of the data.



OdourCollect App compilation process - *OdourCollect*

ENVIRONMENTAL SENSOR DATA

For specificity we would highlight the ATMO Auvergne-Rhône-Alpes data collection station in Grenoble. Here the sensor measures NO_2, $PM_{2.5}$, CO_2, etc. The various gases convey information about the odor of the area under scrutiny. The NO_2 of 85 µg/m³ for example

suggests dense traffic and engine emissions — the smell is likely acrid, burnt, or metallic. The $PM_{2.5}$ of 40 µg/m³ suggests dust or soot, possibly construction or diesel activity. The CO_2 of 550 ppm indicates normal urban air but slight buildup from vehicles and pedestrians. These data from sensors are very relevant for building a smell cartography.



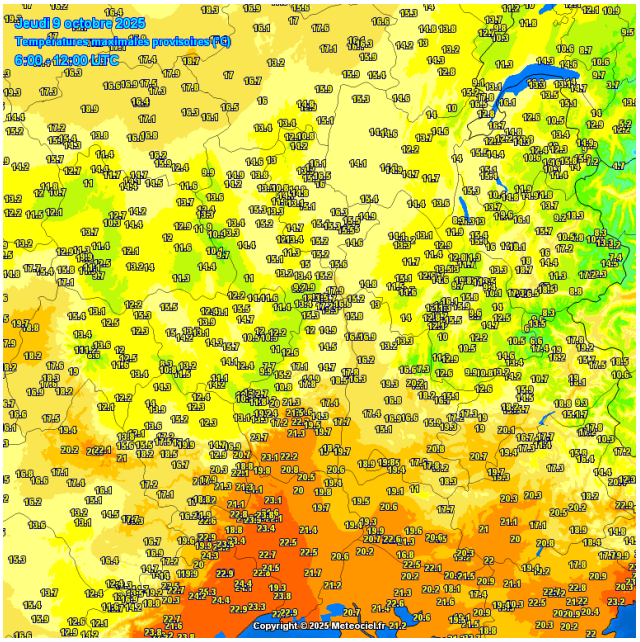
Environmental sensor data collection system - *ATMO Auvergne-Rhône-Alpes*

WEATHER AND CLIMATE DATA

Current APIs provide data on temperature, humidity, and wind based on specific latitude and longitude coordinates. This data can be transformed into weather and climate maps (see example of temperature). When combined with the odors points, this information can support the analysis of dispersion and attenuation of smells.

For example, if a trash odor is detected (through smell reports or a collection point), the high temperatures and humidity can intensify its smell. On the contrary, stronger wind can contribute to the dispersion of these odors, so the intensity will decrease.

The exact factors influencing odor intensity will need to be determined through testing.

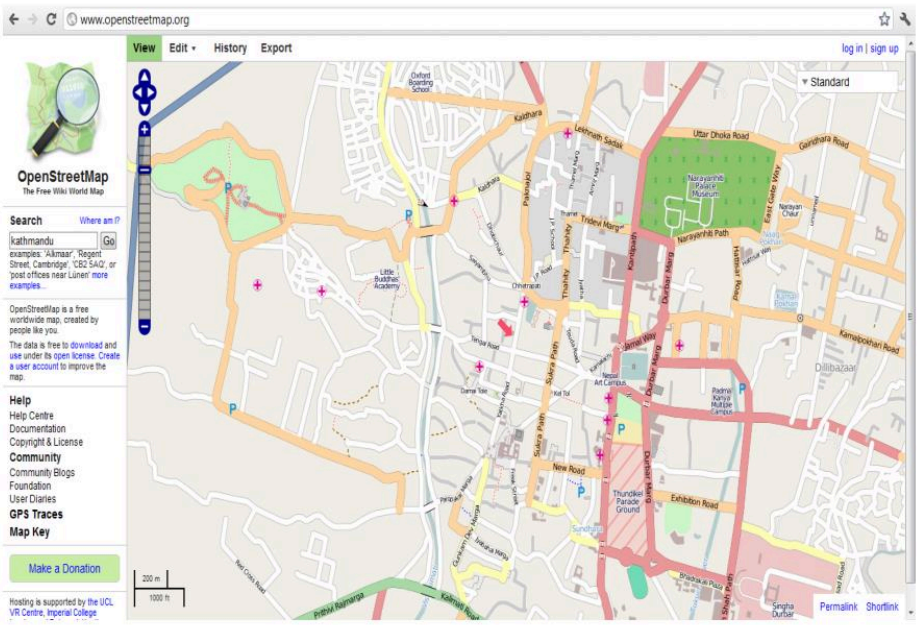


Example of temperature map - *Meteociel*

LOCATION OF RESTAURANTS AND PARKS

This data helps identify scent hotspots — areas where human activity or natural features contribute to a recognizable olfactory “signature.” By utilizing several data sources like the OpenStreetMap. Relevant information is obtained for categorizing these smells, aligning them with specific locations. Some specific examples are - Restaurants / Boulangeries on Rue de Strasbourg, Grenoble Centre. Here, a warm, inviting smell of baked bread and roasted coffee beans lingers along the street - food aromas (bread, coffee, spices, grilled food).

Also, Parks / Green Spaces on Parc Paul Mistral, fresh earthy fragrances mix with floral notes, dispensing floral, earthy, and leafy odors especially after light rain. By tagging accurately these smells and the corresponding locations, relevant data points are established and collected on the apps and these are very important contributors to the smell cartography.

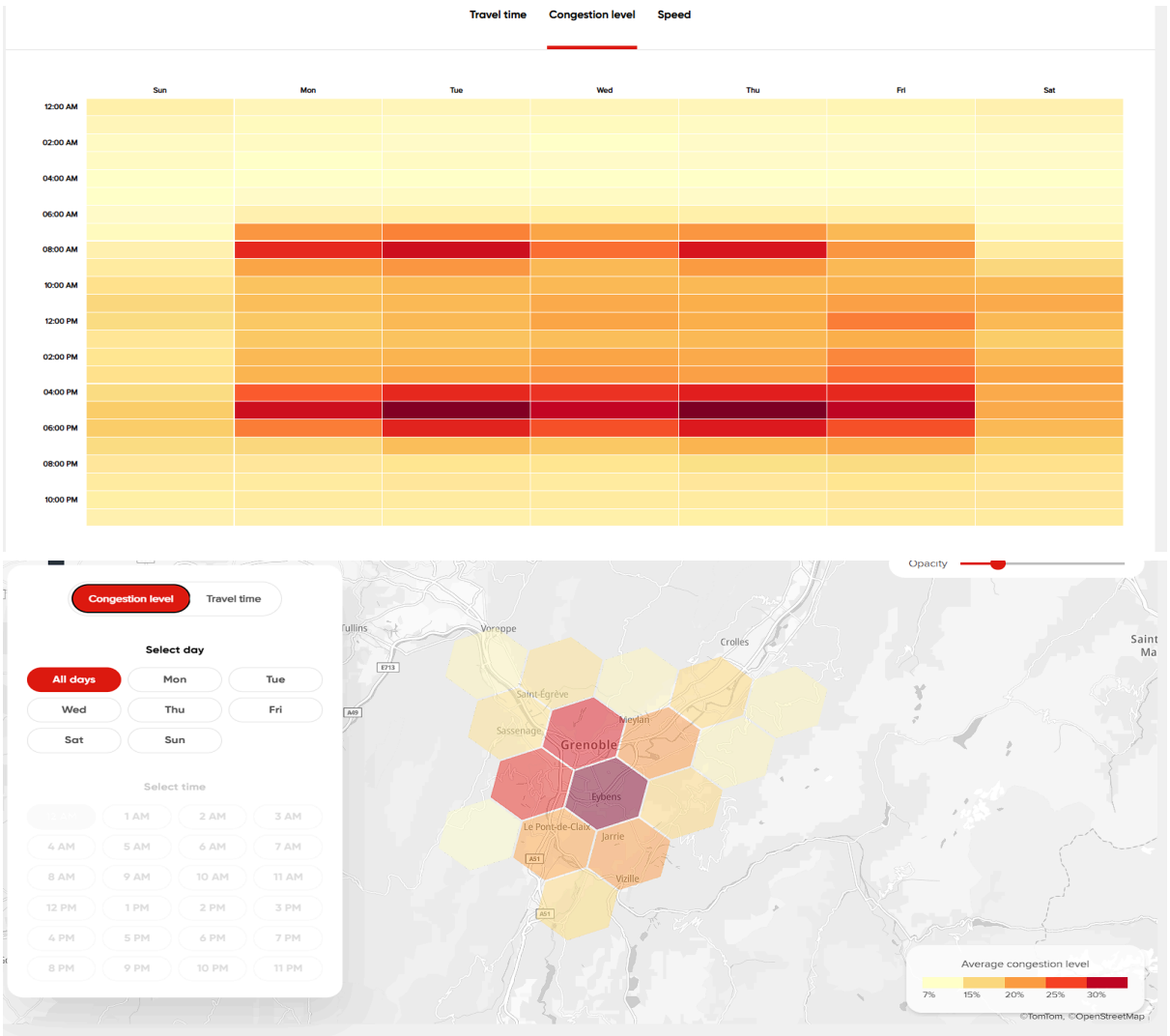


OpenStreetMap section view - *OpenStreetMap*

TRAFFIC

The traffic data type plays a major role regarding the smell and also environmental effects in the city, also knowing the peak traffic hour. Because when we know when the peak hours in the city are, we can easily know that at what times we have more and stronger smells and odors, also it shows when the fuel consumption is at its highest. Also because of the geographical state of the grenoble which it surrounded by mountains when there is no wind to blow away the smokes resulted by the fuel consumption, the smoke and odor will remain still and create a thick cloud of smoke and fume that have a strong smell, and also damaging the environment and also the people that have a breathing disease.

From the data and figures from TomTom Traffic Index, we can see in grenoble, in morning there is a clear sky, and we have 2 traffic pick in the days, one around 8 a.m, other is around 5-7 p.m, that people are going and coming back from their work.



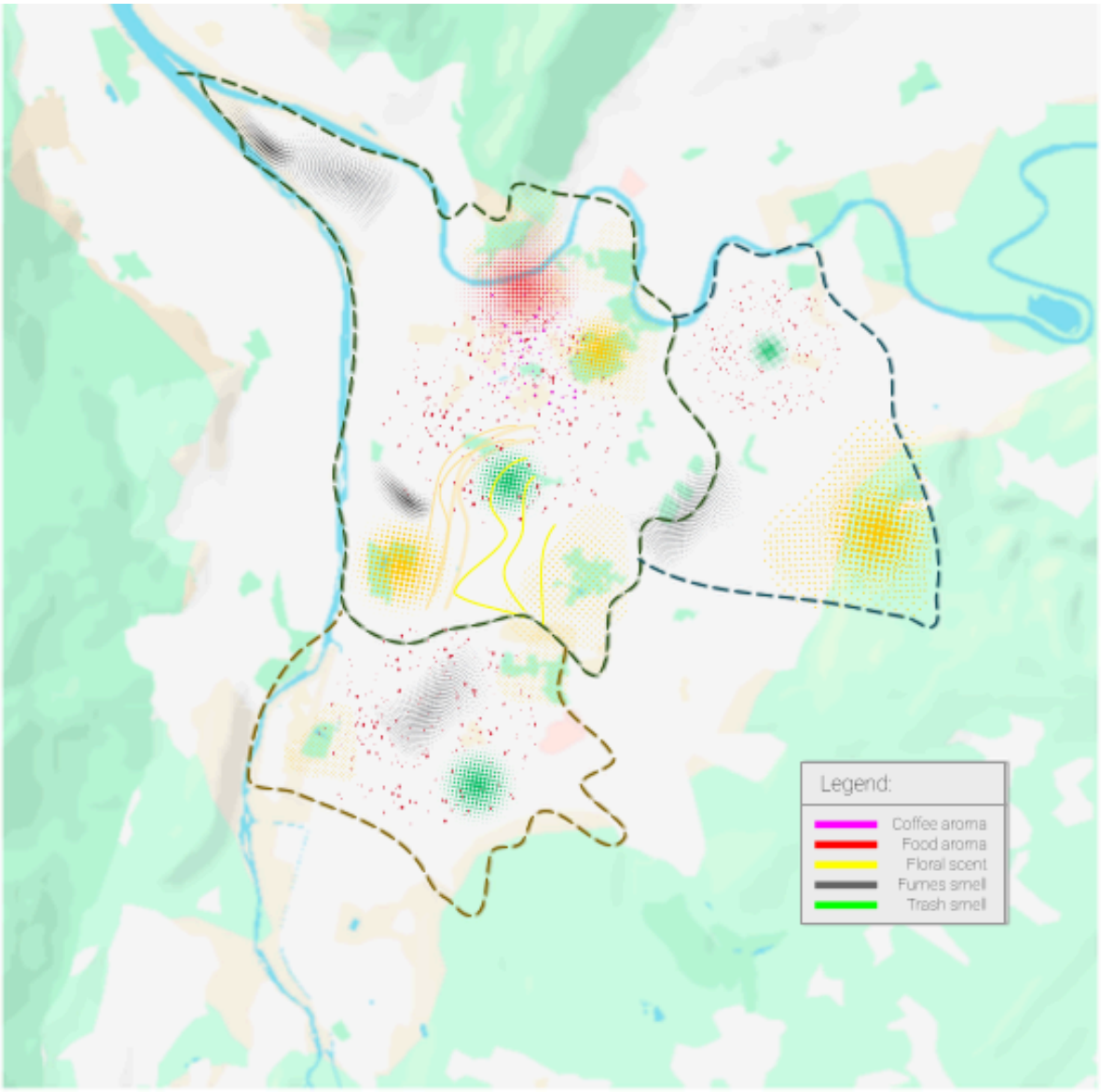
Data of traffic patterns in Grenoble - *TomTom Traffic*

LOCATION OF WASTE DISPOSAL AND COLLECTION POINTS

There is a direct relation between trash locations and trash smells. The data regarding coordinates of the area of the location will help to establish directly the points for trash smells. Similar to the location of restaurants and parks, we will use the areas as high intensity of its respective odor type; for the surrounding area we will decrease the intensity linearly until it reaches zero.

4. Expected result map

In the following image we present an expected results map with the data considered. As we mentioned in the introduction, we are going to give each identify point an intensity and odor type visible by a scale of color.



Smell map of Grenoble, Echirolles, and Saint Martin d'Hères - *Own elaboration*